Robot Vision: Mock Exam 3 (Solutions)

Chapters 2 to 12 · 90 minutes · no documents · 40 points

Part A: Multiple choice

22 points, about 35 min

A1 Why do the hidden layers of a deep network need a nonlinear activation? (ch. 2)

(a) ✓ **Without it, a stack of layers is equivalent to a single affine map.**
(b) It makes the gradient computation cheaper.
(c) It guarantees that the outputs are probabilities.
(d) It reduces the number of parameters.

$W_2(W_1x + b_1) + b_2$ is still affine.

A2 In gradient descent, how are the parameters updated? (ch. 2)

(a) ✓ **They are moved in the direction opposite to the gradient of the loss.**
(b) They are set to the values where the gradient is largest.
(c) They are moved in a random direction.
(d) They are moved along the gradient.

$\mathbf{w} \leftarrow \mathbf{w} - \varepsilon \nabla E$.

A3 How should the mean and standard deviation used to standardize the inputs be computed? (ch. 3)

(a) ✓ **On the training set only, then applied unchanged to the validation and test sets.**
(b) On the test set.
(c) On all the data together.
(d) Separately on each set.

Otherwise information leaks from validation and test.

A4 What is the main difference between AdaGrad and RMSProp? (ch. 3)

(a) RMSProp uses only momentum.
(b) RMSProp accumulates all past squared gradients; AdaGrad uses a moving average.
(c) ✓ **AdaGrad accumulates all past squared gradients; RMSProp uses a moving average of them.**
(d) They are the same method.

AdaGrad's step shrinks forever; RMSProp forgets old gradients.

A5 Which statement relates simple cells, complex cells and CNN operations? (ch. 4)

(a) Complex cells have no receptive field.
(b) Both are like fully connected layers.
(c) Simple cells are like pooling, complex cells like convolution.
(d) ✓ **Simple cells respond to a pattern at a given position (like convolution); complex cells are less sensitive to position (like pooling).**

Hubel and Wiesel; Neocognitron, then LeNet.

A6 Which two properties distinguish a convolutional layer from a fully connected layer? (ch. 4)

(a) Full connections and independent weights.
(b) Full connections and weight sharing.
(c) Local connections and independent weights at each position.
(d) ✓ **Sparse (local) connections and weight sharing across positions.**

Hence far fewer parameters.

A7 In a depthwise separable convolution, what are the roles of the two steps? (ch. 5)

(a) The depthwise convolution mixes channels; the 1×1 handles space.
(b) Both mix channels only.

(c) Both handle only space.
(d) ✓ **The depthwise convolution handles space channel by channel; the 1×1 convolution mixes the channels.**

Used in MobileNet: about $1/D_K^2$ of the cost.

A8 Why can the fully connected layers of a classification CNN be turned into convolutional layers? (ch. 5)

(a) Because pooling can replace them.
(b) Because they always have 1×1 inputs.
(c) Because FC layers have no weights.
(d) ✓ **An FC layer on a $W \times W \times K$ map equals M filters of size $W \times W \times K$, so the network can output a map of predictions.**

This gives the FCN.

A9 What did multifaceted feature visualization show? (ch. 6)

(a) Visualizations do not depend on the initialization.
(b) Each neuron detects exactly one pattern.
(c) Only the first layer can be visualized.
(d) ✓ **A class or a neuron can correspond to several distinct appearances (modes).**

E.g. several views of the same object.

A10 What does CAM require, and how does Grad-CAM remove this requirement? (ch. 6)

(a) CAM needs no special architecture; Grad-CAM needs GAP.
(b) Grad-CAM masks the image with random masks.
(c) ✓ **CAM needs global average pooling followed by a linear classifier; Grad-CAM uses averaged gradients as channel weights.**
(d) Both need a Transformer.

Grad-CAM works on any CNN.

A11 What is the role of reinforcement learning in AutoAugment? (ch. 7)

(a) To train the classifier itself.
(b) To label the augmented images.
(c) ✓ **To search which augmentation operations, probabilities and magnitudes work best.**
(d) To remove bad training images.

A search over policies.

A12 How does FixMatch use weak and strong augmentations? (ch. 7)

(a) Both views are labeled by hand.
(b) The strong view gives the pseudo-label for the weak view.
(c) ✓ **A confident prediction on the weak view becomes the pseudo-label for the strong view.**
(d) Only strong augmentations are used.

The weak view is easier, so more reliable.

A13 What is catastrophic forgetting? (ch. 8)

(a) Dropout removes units permanently.
(b) ✓ **A model loses accuracy on old tasks after learning new ones.**
(c) A model forgets its training data at test time.
(d) The learning rate drops to zero.

The problem of continual learning.

A14 How does Monte Carlo dropout estimate uncertainty? (ch. 8)

(a) ✓ **Dropout stays active at test time and several forward passes with different masks are compared.**
(b) The model is trained several times from scratch.

(c) The softmax temperature is increased.
(d) Dropout is removed and one pass is made.

Spread of the predictions = uncertainty.

A15 Which statement describes the gates of an LSTM? (ch. 9)

(a) All three gates do the same thing.
(b) ✓ **The forget gate controls what memory is kept, the input gate what is added, the output gate what is exposed.**
(c) The output gate erases the memory.
(d) The forget gate adds new information.

Additive memory updates help long-range memory.

A16 In semi-supervised node classification, what is available during training? (ch. 9)

(a) ✓ **The whole graph and all node features, but labels for only some nodes.**
(b) Nothing but the number of nodes.
(c) Only the labeled nodes, without the graph.
(d) Labels for all nodes but no edges.

Structure + features help the unlabeled nodes.

A17 How does attention solve the fixed-length bottleneck of seq2seq? (ch. 10)

(a) It removes the decoder.
(b) It uses only the last encoder state.
(c) ✓ **At each decoding step, it computes a new weighted combination of all encoder states.**
(d) It makes the hidden state longer.

The weights change at each output step.

A18 How is the output for one query computed in scaled dot-product attention? (ch. 10)

(a) ✓ **The query is compared with all keys, the scores go through a softmax, and the weights combine the values.**
(b) Only the most similar key is kept.
(c) The keys are added to the query.
(d) The query is compared with all values, and the keys are averaged.

$\text{softmax}(QK^\top/\sqrt{d_k})V$.

A19 In cross-attention for VQA, where do the queries, keys and values come from? (ch. 11)

(a) ✓ **Queries from the question tokens; keys and values from the image tokens.**
(b) All three from the image tokens.
(c) All three from the question tokens.
(d) Queries from the image; keys from the answer.

Self-attention: all from the same sequence.

A20 In an LLM agent, where does a tool call (search, code) run? (ch. 11)

(a) ✓ **Outside the LLM; its result comes back as a new observation.**
(b) In the LLM's chain of thought.
(c) Nowhere: the LLM imagines the result.
(d) Inside the LLM's weights.

Observe, decide, act.

A21 In a GAN, what are the goals of the generator G and the discriminator D? (ch. 12)

(a) Both try to classify the real images.
(b) G tries to tell real from fake; D generates images.
(c) ✓ **G tries to produce images that D takes for real; D tries to tell real images from generated ones.**

(d) G reconstructs its input; D predicts noise.

A min-max game; no likelihood.

A22 What is special about a normalizing flow? (ch. 12)

(a) It generates by many denoising steps.
(b) It needs a discriminator.
(c) It only gives a lower bound on the likelihood.
(d) **✓ It uses an invertible map, so the exact density of an observation can be computed by change of variables.**

Trained by maximum likelihood.

Part B: Short answers

12 points, 3 per question, about 35 min

B1 Give two signs of overfitting and two ways to reduce it. (ch. 3)

Training error keeps falling while validation error rises; a large gap between them. Remedies: more data or augmentation, weight decay, dropout, early stopping.

B2 Why does SSD predict from feature maps of several resolutions, and what is NMS used for? (ch. 5)

Fine maps detect small objects, coarse maps large ones. NMS removes duplicate boxes that cover the same object.

B3 In multiple instance learning, why must the aggregation function be permutation-invariant? Give one example. (ch. 7)

The instances of a bag have no natural order, so the bag output must not depend on it. Examples: max, mean, or attention pooling.

B4 What is the difference between training and sampling in a diffusion model? (ch. 12)

Training samples one random time step per update and learns to predict the added noise. Sampling starts from pure noise and runs all the reverse steps in sequence, which is slower.

Part C: Calculations

6 points, 2 per question, about 15 min

C1 Three 3×3 convolutions with stride 1 are stacked. What is the receptive field of an output unit? (ch. 4)

$1 + 3 \times (3 - 1) = \mathbf{7}$, i.e. 7×7.

C2 A causal TCN has kernel size 2 and four layers with dilations 1, 2, 4, 8. What is its receptive field? (ch. 9)

$1 + (2 - 1)(1 + 2 + 4 + 8) = \mathbf{16}$ time steps.

C3 How many weights (no bias) does a 3×3 convolution from 64 to 128 channels have as a standard convolution, and as a depthwise separable one? (ch. 5)

Standard: $3 \cdot 3 \cdot 64 \cdot 128 = \mathbf{73\,728}$. Separable: $3 \cdot 3 \cdot 64 + 64 \cdot 128 = 576 + 8192 = \mathbf{8768}$ (about $8\times$ fewer).